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Editor-in-Chief
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Dear Editor,

We submit for your consideration:

**Wide-Area Consensus via GOOSE for Coordinated Protection in IBR-Rich
Distribution Networks**

The problem. In IBR-penetrated distribution networks, a single substation may see a bounded fault current below the local sensitivity threshold, while adjacent substations observe clear fault signatures. No individual relay has the full picture. Classical wide-area protection schemes aggregate raw measurements but do not account for the varying *reliability* of estimates across substations under IBR operating conditions.

Contributions.

1. **Quality-weighted consensus (C14):** Trip votes are weighted by $w_i = \max(0, 1 - k_n^{(i)}/30)$, derived directly from the SAMBPS condition number $k_n^{(i)}$. Ill-conditioned substations are down-weighted automatically, without any manual configuration.
2. **Conservativeness preservation (Theorem 1):** The multi-node consensus trip rule is proven to inherit the zero-false-trip guarantee of the single-node CUEP trip rule (C13), making the aggregate at least as conservative as any constituent relay.
3. **Dropout robustness (Proposition 1):** Zone accuracy $\geq 94.8\%$ with up to $\lfloor (N_{SS} - 1)/2 \rfloor$ simultaneous substation dropouts, degrading gracefully to unweighted majority in the limit.
4. **Hardware validation:** 6.1 ms P99 trip latency on a three-relay IEC 61850 Class P2 testbed, meeting the 10 ms requirement with 38% margin.

Fit to *IEEE Transactions on Smart Grid*. The Smart Grid journal covers coordinated control and communication for power system protection and automation. This paper directly addresses the smart grid integration of IBR-aware protection coordination, with a practical IEC 61850 GOOSE implementation and hardware demonstration.

Suggested reviewers.

- Prof. Arun G. Phadke (Virginia Tech) — wide-area protection
- Prof. Ali Hooshyar (University of Toronto) — IBR protection
- Prof. Mladen Kezunovic (Texas A&M) — smart grid relaying
- Dr. Alexander Apostolov (OMICRON) — IEC 61850, GOOSE

This manuscript is not under review elsewhere. All authors have approved the submission.

Sincerely,

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